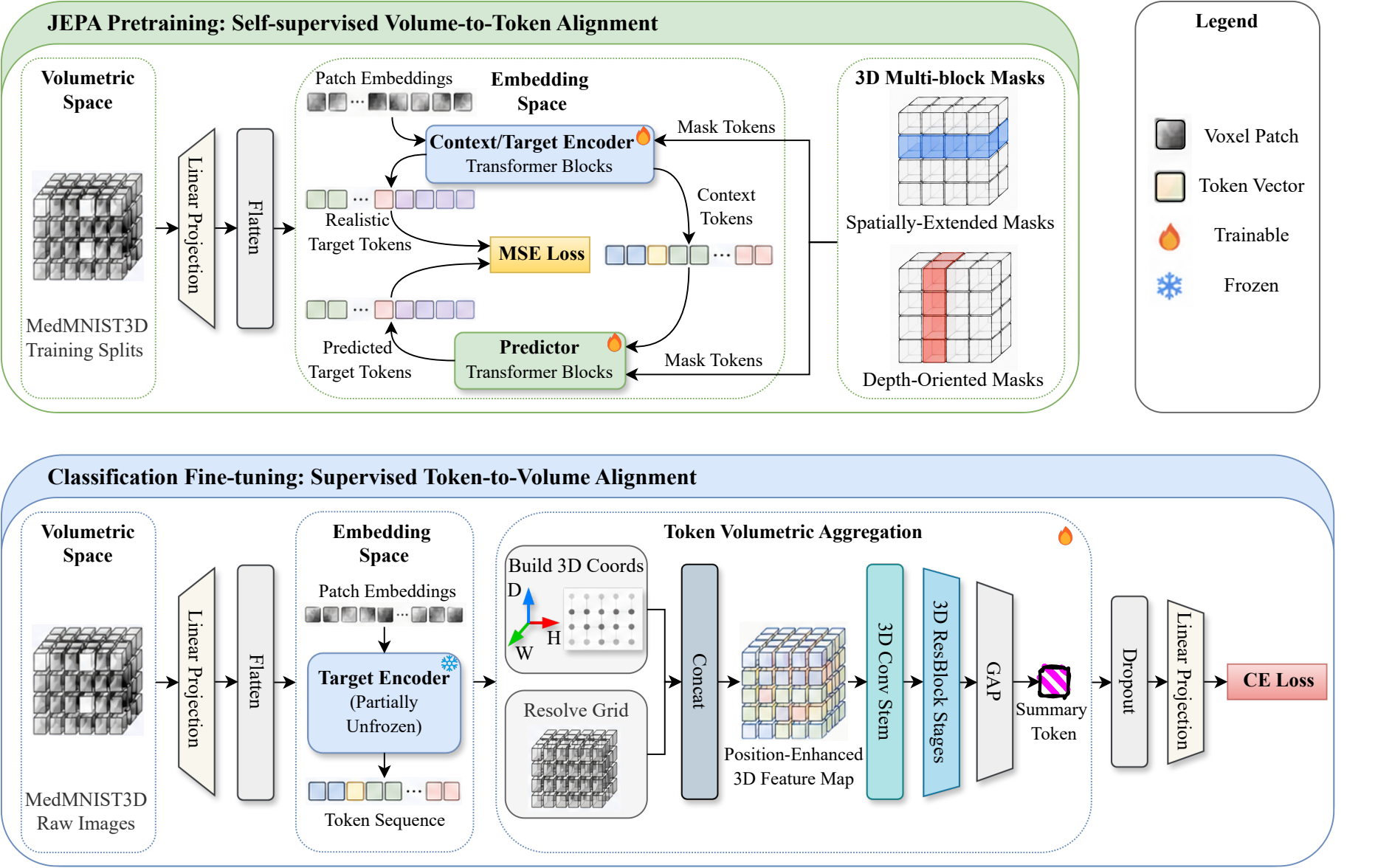

JEPA Pretraining: Self-supervised Volume-to-Token Alignment
Volumetric Space
MedMNIST3D Training Splits
Linear Projection
Flatten
Patch Embeddings
Embedding Space
Context/Target Encoder
Transformer Blocks
Mask Tokens
Context Tokens
Realistic Target Tokens
MSE Loss
Predicted Target Tokens
Predictor
Transformer Blocks
Mask Tokens
3D Multi-block Masks
Spatially-Extended Masks
Depth-Oriented Masks
Legend
Voxel Patch
Token Vector
Trainable
Frozen
Classification Fine-tuning: Supervised Token-to-Volume Alignment
Volumetric Space
MedMNIST3D Raw Images
Linear Projection
Flatten
Embedding Space
Patch Embeddings
Target Encoder
(Partially Unfrozen)
Token Sequence
Token Volumetric Aggregation
Build 3D Coords
D
H
W
Resolve Grid
Concat
Position-Enhanced 3D Feature Map
3D Conv Stem
3D ResBlock Stages
GAP
Summary Token
Dropout
Linear Projection
CE Loss